Quantification: 543-2
==========
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---
measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
---
measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	9.1667	9	2.8868	count
ring_length_r30_px	headline	414.75	391	93.4395	px
ring_length_r60_px	reported	1115.08	1113.5	223.058	px
owned_length_px	ownership-dependent	3605.71	3584.34	1493.78	px
edge_count	ownership-dependent	169.75	165.5	68.5872	count
mean_edge_length_px	ownership-dependent	21.0798	21.0256	1.6856	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0472403	0.0466671	0.00668715	px^-1
ring_density_r60	normalised	0.0500196	0.0514351	0.00588809	px^-1
roots_per_100px_perimeter	normalised	4.0724	3.9619	0.7291	per 100 px
ring_attached_length_r30_px	network v2	375.417	352	104.821	px
ring_attached_length_r60_px	network v2	1017.5	973.5	226.887	px
ring_length_w_r30_px	network v2	468.713	441.933	108.156	px
ring_length_w_r60_px	network v2	1274.26	1267.77	256.362	px
sholl_crossings_r10	network v2	11	11	2.9848	count
sholl_crossings_r20	network v2	13.5833	12.5	2.7455	count
sholl_crossings_r30	network v2	14.9167	15	3.4234	count
loop_count	loops, report only	20	19	10.1802	count
loop_edge_share	loops, report only	0.4412	0.4501	0.0908	share
ring_width_mean_r30_px	width	7.3776	7.3593	0.3037	px

Per field
---
measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	44785	px
canalicular_length_density_per_px	0.0437251	px^-1
junction_count	1160	count
junction_density_per_px2	0.00113255	px^-2
lacuna_count	12	count

lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	146	count
mean_component_length_px	306.747	px
median_component_length_px	16.5	px
field_length_density_w_per_px	0.0498039	px^-1
field_density_without_flagged_per_px	0.0437251	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	416	px
unattached_component_count	24	count
canalicular_length_density_attached_per_px	0.0437251	px^-1
field_length_density_with_unattached_per_px	0.0441312	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	285	count
loop_edge_share	0.4972	share
edge_count	2305	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9948	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	44785	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	13	514	1397	4203.44	207	20.3065	7.8391
2	13	639	1518	5640.65	276	20.4371	6.9516
3	8	395	981	2355.89	109	21.6137	7.5509
4	9	382	710	1223.9	71	17.2381	7.0693
5	6	355	961	2569.59	127	20.233	7.1713

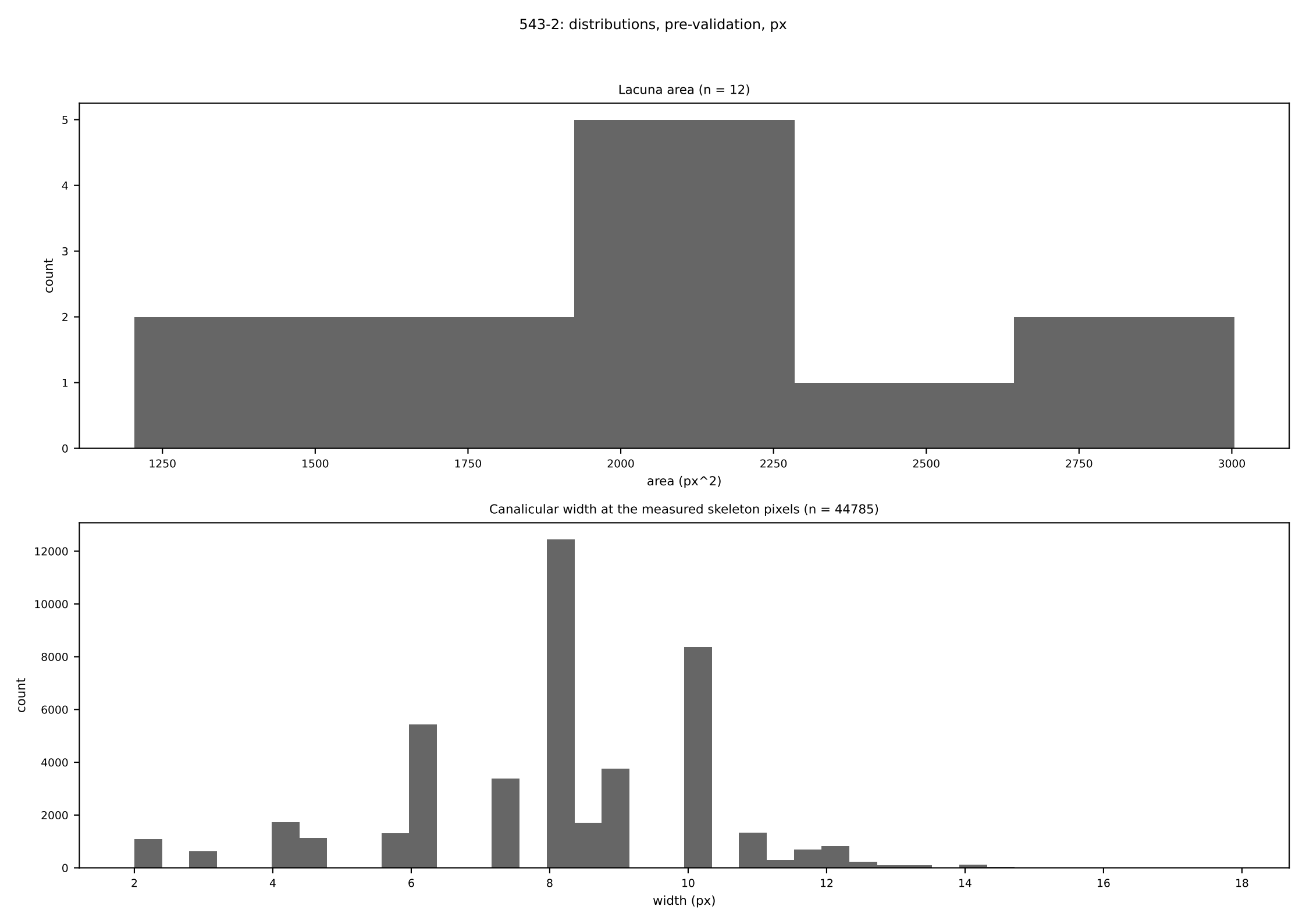
6	10	354	1023	1830.28	88	20.7986	7.4527
7	5	301	938	3080.09	126	24.4451	7.1774
8	13	469	1226	4088.59	191	21.4062	7.4201
9	11	424	1229	5798.87	265	21.8825	7.4156
10	9	387	1204	4739.32	223	21.2526	7.9823
11	7	441	1211	4828.9	214	22.565	7.1985
12	6	316	983	2909.04	140	20.7789	7.3029

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0540484	0.0587197	5.3171
2	295.12	10519	26271	1	1	0.0607472	0.0577824	4.405
3	172.894	7548	18977	1	1	0.0523317	0.0516942	4.6271
4	232.551	8756	18925	0.9575	0.7928	0.0436272	0.0375165	3.8701
5	154.267	7010	17818	1	1	0.0506419	0.0539342	3.8894
6	231.764	9080	22988	1	1	0.0389868	0.0445015	4.3147
7	155.924	7017	19168	1	1	0.0428958	0.0489357	3.2067
8	254.25	9429	23720	1	1	0.0497402	0.0516863	5.1131
9	277.078	10094	25807	1	1	0.0420051	0.0476227	3.97
10	227.622	8939	23523	1	1	0.0432934	0.0511839	3.9539
11	225.522	8872	23390	1	1	0.0497069	0.0517743	3.1039
12	193.723	8132	21901	1	1	0.0388588	0.0448838	3.0972

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	485	1329	587.588	1609.02	13	16	16
2	632	1479	725.357	1731.02	17	20	21
3	363	807	444.262	1125.69	12	14	14
4	356	647	429.534	793.955	12	13	8
5	276	896	396.836	1111.18	8	11	15
6	335	1001	403.706	1172.76	10	11	11
7	251	902	338.108	1072.75	7	11	12
8	452	1082	539.103	1401.13	14	12	17
9	348	893	469.877	1407.8	9	16	15
10	306	1152	439.605	1362.77	8	12	18
11	388	1076	497.818	1372.3	13	15	17
12	313	946	352.764	1130.76	9	12	15


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44785)
count
width (px)